Atom Vayalinkal

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Education

Economics Ph.D., University of Toronto — 2021–Present

Honours Bachelor of Science, University of Toronto — 2015–2019
Economics (major), Mathematics and Statistics (minors)

Research

Job Market Paper

Many Alternatives, Random Incentives: Sharp Bounds and Efficiency in Treatment Selection Models

Abstract: In this paper, I present new results on partial-identification and specification-testing in models with essential heterogeneity, (as-if) randomly assigned discrete instruments, discrete treatments, and general outcomes. I consider a broad class of models that restrict how instruments influence selection into treatment while remaining agnostic about outcomes, including, as special cases, *all* models that simply rule out certain ways of responding to the instrument. I characterize their sharp testable implications as convex moment inequalities and, for a broad and flexible class of target parameters, express sharp bounds and parameter-inclusion tests as solutions to finite-dimensional convex programs; several cases are also shown to admit analytic sharp bounds. These results follow from a finite-dimensional convex representation of such models' empirical content, introduced and developed in this paper, with key support functions computable exactly from finitely many linear programs even when outcomes are continuous. The resulting framework enables statistical inference using existing methods, without discretizing outcomes. I also derive a semiparametric efficiency bound for estimation of the sharp bounds and propose a cross-fitted bias-corrected estimator that asymptotically attains this bound in suitably regular cases, while mitigating the inward bias associated with analog estimators in irregular cases. Simulations show that the proposed methodology can yield substantially more informative conclusions than discretization-based approaches, often at much lower computational cost.

Publications

Multiple Testing with Covariate Adjustment in Experimental Economics
(with John A. List and Azeem M. Shaikh). *Journal of Applied Econometrics*, May 2023.

Abstract. This paper provides a framework for testing multiple null hypotheses simultaneously using experimental data in which simple random sampling is used to assign treatment status to units. Using general results from the multiple testing literature, we develop under weak assumptions a procedure that (i) asymptotically controls the familywise error ratethe probability of one or more false rejectionsand (ii) is asymptotically balanced in that the marginal probability of rejecting any true null hypothesis is approximately equal in large samples. Our procedure improves upon classical methods by incorporating information about the joint dependence structure of the test statistics when determining which null hypotheses to reject, leading to gains in power. An important point of departure from prior

work is that we exploit observed, baseline covariates to obtain further gains in power. The way in which we incorporate these covariates is based on recent results from the statistics literature in order to ensure that inferences are typically more powerful in large samples.

Working Papers

Horowitz-Manski-Lee Bounds with Multilayered Sample Selection
(with Kory Kroft and Ismael Mourifié). *NBER Working Paper No. 32952*, June 2026.

Abstract. This paper studies partial identification of treatment effects in the presence of sample selection, where treatment affects both selection into the sample and sorting across layers with heterogeneous outcomes. We show that canonical Lee bounds identify a total effect that combines the within-layer causal effect of treatment with a sorting effect reflecting outcome differences across layers. We derive sharp bounds on the within-layer causal effect using a two-step approach that extends Horowitz and Manski (1995) to a system of mixture equations with cross-equation dependence. Further, we show that under additional restrictions, these within-layer effects are suffcient for welfare analysis. Two empirical applications to job training experiments illustrate the approach; our estimates show that even when Lee bounds are strictly positive, within-firm bounds can be tight around zero, suggesting that Lee bounds capture a pure sorting effect.

Nothing Propinks Like Propinquity: Using Machine Learning to Estimate the Effects of Spatial Proximity in the Major League Baseball Draft
(with Majid Ahmadi, Nathan Durst, Jeff Lachman, John A. List, Mason List, and Noah List). *NBER Working Paper No. 30786*, December 2022.

Abstract. Recent models and empirical work on network formation emphasize the importance of propinquity in producing strong interpersonal connections. Yet, one might wonder how deep such insights run, as thus far empirical results rely on survey and lab-based evidence. In this study, we examine propinquity in a high-stakes setting of talent allocation: the Major League Baseball (MLB) Draft from 2000-2019 (30,000 players were drafted from a player pool of more than a million potential draftees). Our findings can be summarized in four parts. First, propinquity is alive and well in our setting, and spans even the latter years of our sample, when higher-level statistical exercises have become the norm rather than the exception. Second, the measured effect size is consequential, as MLB clubs pay a significant opportunity cost in terms of inferior talent acquired due to propinquity bias: for example, their draft picks are 38% less likely to ever play a MLB game relative to players drafted without propinquity bias. Third, those players who benefit from propinquity bias fare better both in terms of the timing of their draft picks and their initial financial contract, conditional on draft order. Finally, the effect is found to be the most pronounced in later rounds of the draft, where the Scouting Director has the greatest latitude.

Work in Progress

Multiple Testing under Covariate-Adaptive Randomization
(with Azeem M. Shaikh). *Draft available upon request.*

Abstract. In randomized experiments with multiple comparison groups, where multiple hy-

pothesis testing is a natural concern, it is often desirable to adjust for covariates at both the design and analysis stages. Covariate-adaptive randomization is widely used to ensure balance in treatment allocation across important prognostic covariates, and covariate adjustment is often used to improve efficiency while producing asymptotically valid inference. This paper extends the multiple testing procedure of List, Shaikh and Vayalinkal (2023), which enables balanced control of the familywise error rate under simple randomization, to experiments conducted under covariate-adaptive randomization. The resulting framework enables joint hypothesis testing for multiple treatments, simultaneous inference about treatment effects on multiple outcomes, and inference on heterogeneous treatment effects through subgroup analysis.

Monitoring Bureaucrats: Evaluating the Dynamic Impact of Audits under Contingent Measurement
(with Gustavo Bobonis and Ismael Mourifié). *In Progress.*

Honours & Awards

Ontario Graduate Scholarship ($10,000) 2026
SSHRC Canadian Graduate Scholarship - Doctoral ($110,000) 2022–2025
Best Paper Award (Methods), Chicago Causal Inference Student Conference 2024

Other Research and Teaching Experience

Research Assistance

Pre-Doctoral Research Assistant, University of Chicago 2019–2021
Chicago Experiments Group, Becker Friedman Institute. Supervisor: John List.

Teaching Assistance

University of Toronto

ECO2010: Mathematics and Statistics Course for PhD Students 2023, 2024, 2026
ECO439: Empirical Methods in Microeconomics 2025-2026
ECO1010: Mathematics and Statistics Course for Master's Students 2023-2025
ECO403: Topics in Development Economics 2025
ECO358: Financial Economics I 2025
ECO475: Applied Econometrics II 2022, 2023
ECO333: Urban Economics 2021-2023
ECO1200: Microeconomic Theory (for Master of Financial Economics) 2023
ECO404: Topics in Managerial Economics 2023
ECO375: Applied Econometrics I 2023
ECO208: Macroeconomic Theory 2022
ECO204: Microeconomic Theory and Applications (for Commerce) 2023, 2024

Programming Stata, Mata, Mathematica, Python, R, JavaScript, SQL, PHP

Citizenship Canadian